SecureGuard: Risk-Based Adaptive Security for MPIN Reset

Challenge 6: Risk-Based Security for MPIN Reset

May 10, 2026

Team Member Details

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Problem Understanding

The Challenge:

- MPIN reset is a critical vulnerability.
- Fraudsters exploit OTPs via social engineering and scripts.

Why It Matters:

- Current systems lack **context**.
- Users on suspicious calls or compromised devices are treated identically to safe users.

Current Gap

No behavior-based risk scoring occurs **before** the OTP is displayed.

Proposed Solution: SecureGuard

Core Concept: Real-time risk evaluation middleware.

- **Risk Scoring Engine:** Calculates Low, Mid, or High risk scores in milliseconds based on user context.
- **Adaptive UI:** Instead of a static OTP screen, the interface changes dynamically:

1. **Stealth Mode:** Zero friction for safe users.
2. **Caution Mode:** Yellow warnings to build awareness.
3. **Intervention Mode:** Full friction (timers/checks) to stop fraud.

Key Features

1. Signal Detection

- High Value / New Payee flags.
- Unusual time detection.
- Active call status check.
- Device integrity scan.

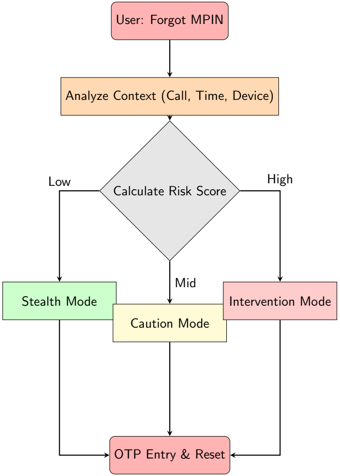
2. Risk Calculation

- Runs real-time (before OTP).
- Outputs Low / Mid / High score.

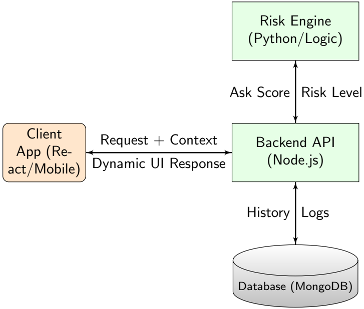
3. Adaptive Response

- Dynamic Warning Messages.
- (e.g., "Don't share OTP")
- Interactive friction (Timers).

User Journey / Flow



System Architecture



Technology Stack & Scope

Tech Stack

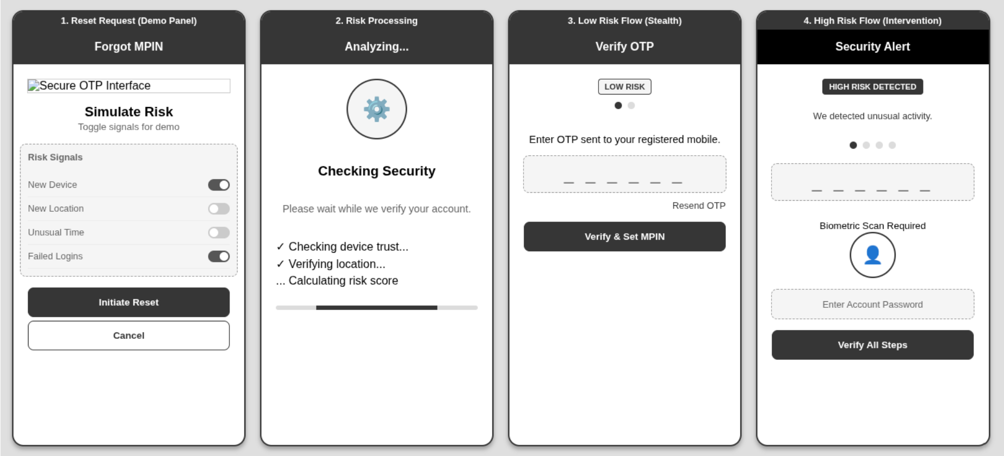
- **Frontend:** React.js / Flutter
- **Backend:** Node.js (Express)
- **Database:** MongoDB
- **Logic:** Python (Simulation)

Scope

- **In Scope:** Risk eval, Active call detection, Dynamic UI.
- **Out of Scope:** Biometrics, Full core banking logic.

UI/UX Prototypes: Secure OTP Interface

Secure OTP Interface



Clean OTP display with prominent security warning.

Expected Impact & Conclusion

Impact:

- Stops fraud scripts by forcing user attention (Intervention).
- Protects against Vishing (Call detection).
- Zero friction for 90% of legitimate users (Stealth mode).

Thank You!

Questions? Github Link: https://github.com/sagarEmn/esewa-hackathon-dynamic-otp-ui